

ICT1011: Computer Organisation and Architecture

Tutorial 8: Secondary Memory Subsystems

Q1. Magnetic Memory

1. A magnetic hard disk has 8 recording surfaces with a total of 6,000 cylinders. There is an average of 200 sectors per track and each sector contains 512 bytes of data. The hard disk operates with a rotational speed of 7200 rpm and has a 4 ms average seek time (or also known as seek delay). The track-to-track access time is 2.5 ms.
 - a. What is the capacity of this hard disk (in bytes)?
 - b. What is the average data transfer rate (in bytes per second) for accessing consecutive data blocks on the same track?
 - c. Assume the average size of a block being accessed is 64Kbytes and each data block is stored in consecutive sectors. What is total time required to access a random block of data on the hard disk?
 - d. Assuming that another data block can be striped to exactly 6 cylinders:
 - 1) What is the total time required to transfer the whole data block? Assume that data on consecutive tracks within a cylinder is read sequentially without additional seek time, and is stored continuously across tracks.
 - 2) Now, consider data striping by platter. What is the total time required to transfer the whole data block? (Data is stored in consecutive tracks and sectors)
 - 3) Compare the above results and illustrate the advantage of data striping by cylinder.

Q2. RAID

2. Consider a RAID array consisting of 6 drives, each with a capacity of 100GB.
 - a. Determine the available data storage capacity for RAID 0, RAID 1, and RAID 10, respectively.
 - b. Suppose there is a single 60GB data file that is divided into 6 blocks (BLK1-6), with each block having a size of 10GB. Provide a visual representation of how this file is stored in a RAID 10 configuration.